DAY 2 · FRIDAY 23 OCTOBER



Agents I

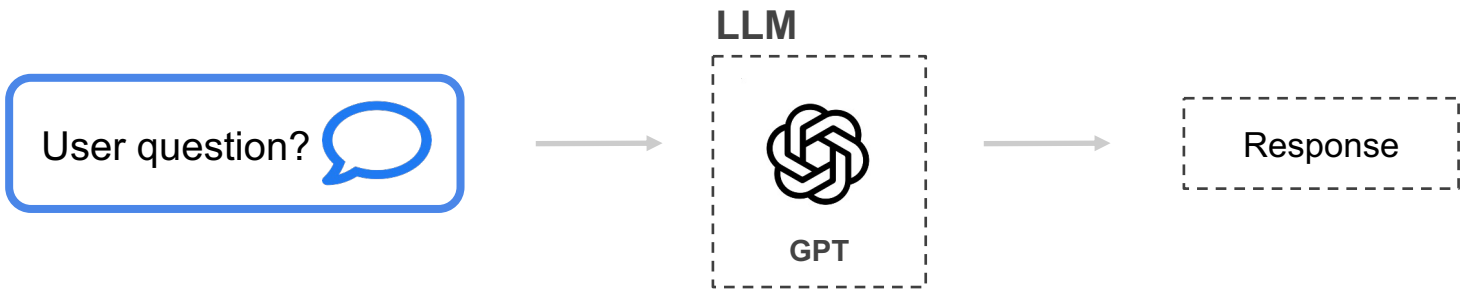
Reasoning models, the agent loop, tools and MCP

Pavlos Protopapas & Shivas Jayaram
Harvard University

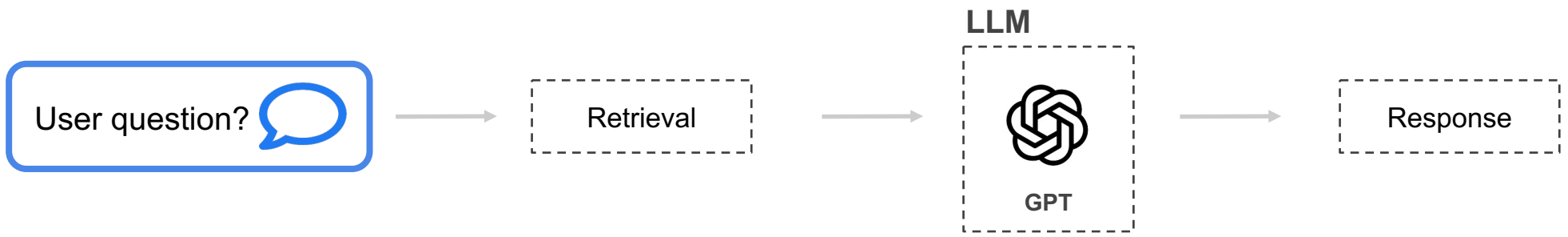
Outline

1. Recap
2. Reasoning Models
3. Motivation
4. Agents
5. Tools, MCP
6. Tutorials

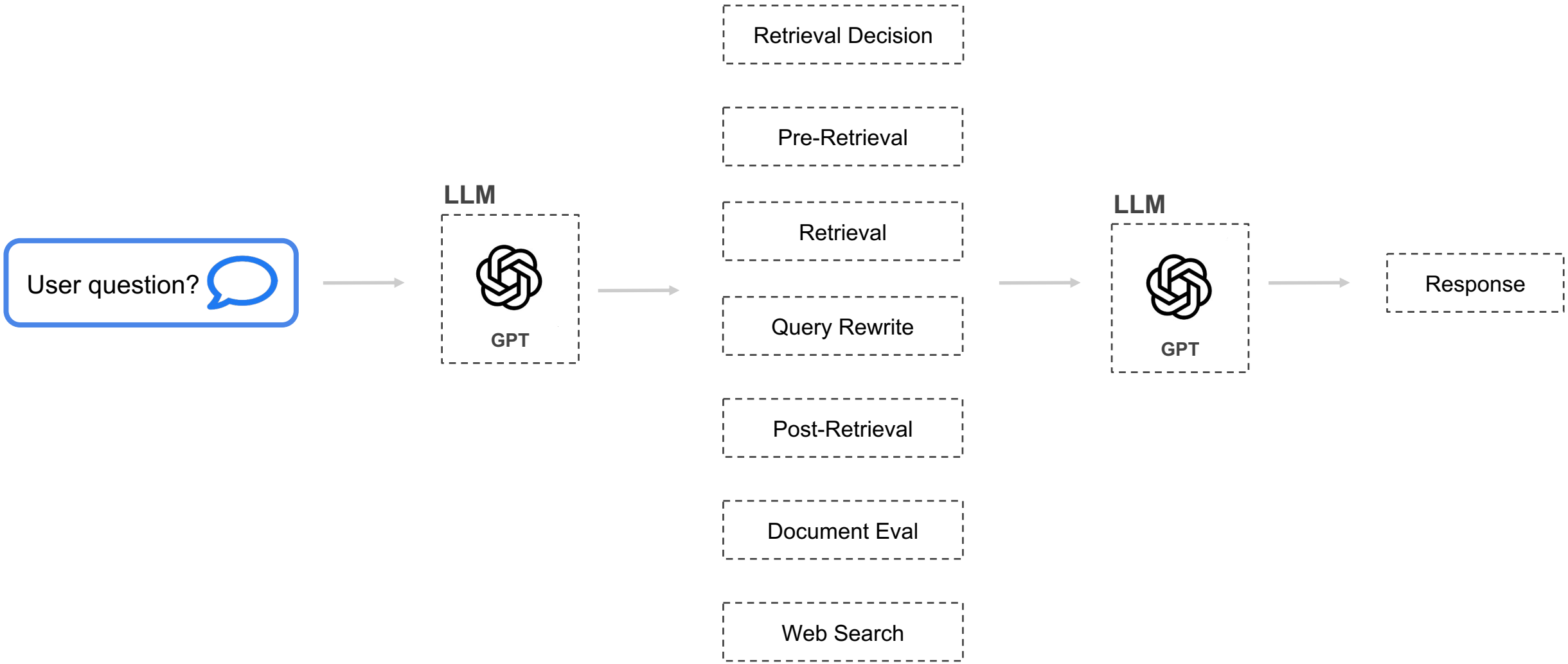
Recap: LLM



Recap: Naive RAG



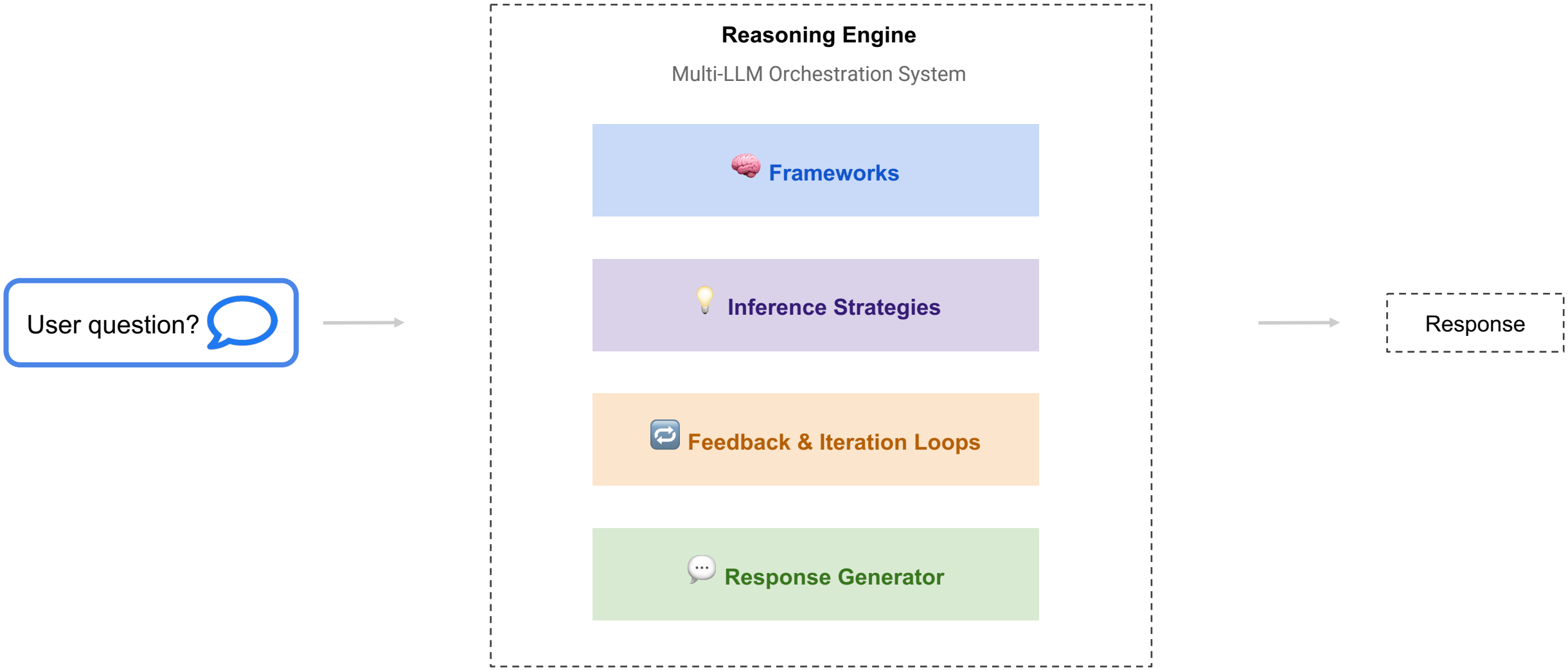
Recap: Agentic RAG



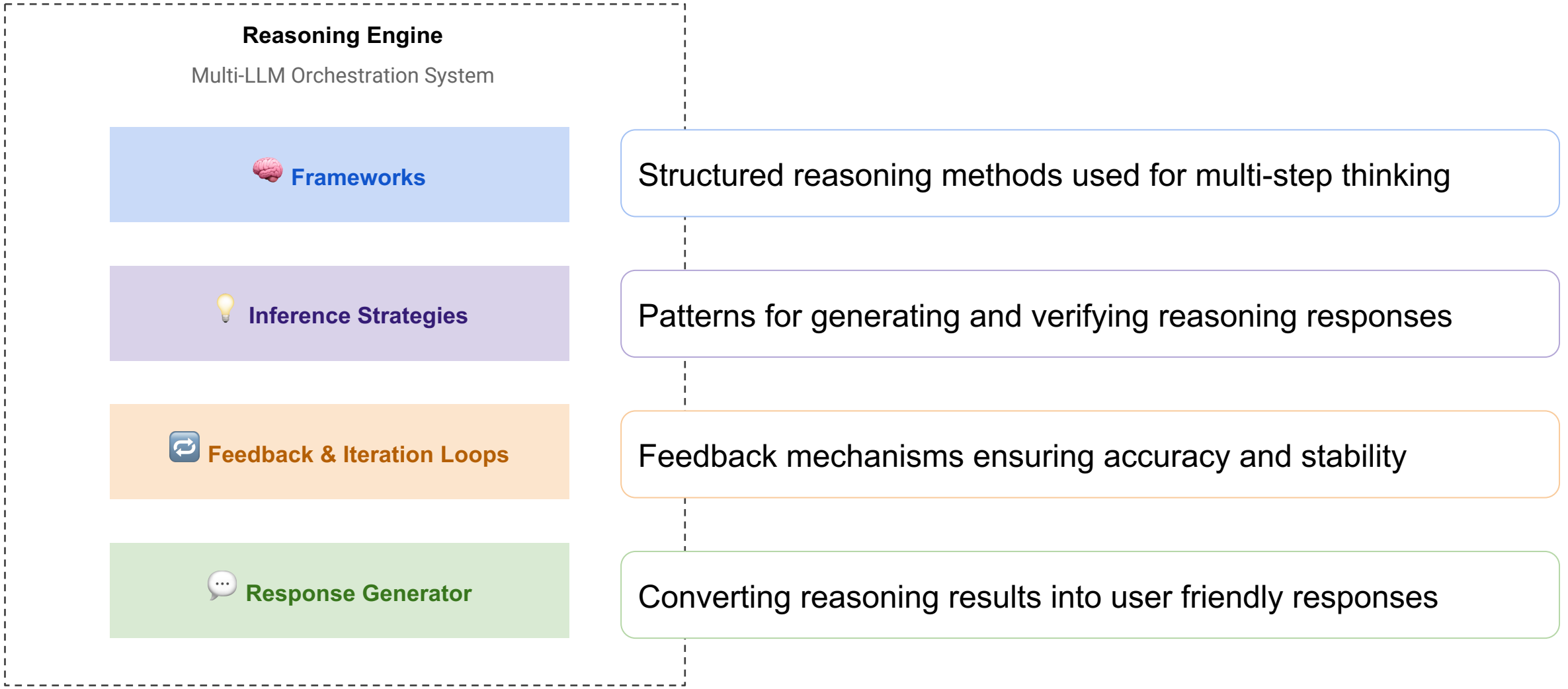
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2. **Reasoning Models**
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Reasoning LLM



Reasoning LLM



Reasoning LLM

🧠 Frameworks

- Chain of Thought (COT)
- Problem Decomposition
- Tree of Thought (ToT)
- Graph of Thought (GoT)
- Self-Refinement

💡 Inference Strategies

- Parallel Decoding
- Sequential Processing
- Majority Voting / Ensembles
- Rationale Verification
- Backward Chaining
- Iterative Prompting

🔁 Feedback & Iteration Loops

- Self-Consistency Loop
- Decomposition Loop
- Self-Refinement Loop
- Verification Loop
- Dynamic Scaling
- Bootstrapping Loop

💬 Response Generator

- Summary Generation
- Answer Formatting
- UI adaptation
- Confidence Scoring
- Multi-Modal Output
- Error Detection & Fallback

Outline

1. Recap
2. Reasoning Models
3. **Motivation**
4. Agents
5. Context Engineering
6. Tools, MCP
7. Agent Systems
8. Additional Resources

Motivation: Travel Planner

Let's say we pass the following prompt to the LLM

I want to plan a 5-day trip to Paris.

The LLM should help with the following:

1. Find flights and hotels within our budget.
2. Suggest activities like visiting the Louvre etc, tailored to our preferences.
3. Dynamically adjust plans if there's a weather issue or a flight delay.

A seamless travel experience where everything is taken care of effortlessly.

Motivation: Travel Planner

Can RAGs or LLMs alone achieve this?

Let's say we pass t

Paris.

The LLM should help with the following:

1 budget.

2 e Louvre etc, ta

3 e's a weather is

A seamless travel experience where everything is taken care of effortlessly.

LLMs

Can generate itineraries but lack **dynamic decision-making** capabilities.

RAGs

Limited to **retrieving and summarizing information** (e.g., fetching flight options).

Motivation: Travel Planner

So, how do we implement this travel planner? What do LLMs and RAG lack?

1. **No Interaction with the External World:**
   Cannot access real-time systems like weather, maps, or booking platforms.
2. **Absence of Tools:**
   Lacks the ability to directly use APIs or services to book flights, hotels, or other essentials.
3. **Lack of Autonomy:**
   Cannot make decisions or take actions independently to adapt to changing situations.

We need something more—an advanced system that can interact, adapt, and act.

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Agents

What is an Agent?

Agent Definition

*"Agents are **systems** that independently accomplish tasks on your behalf."* ~ OpenAI

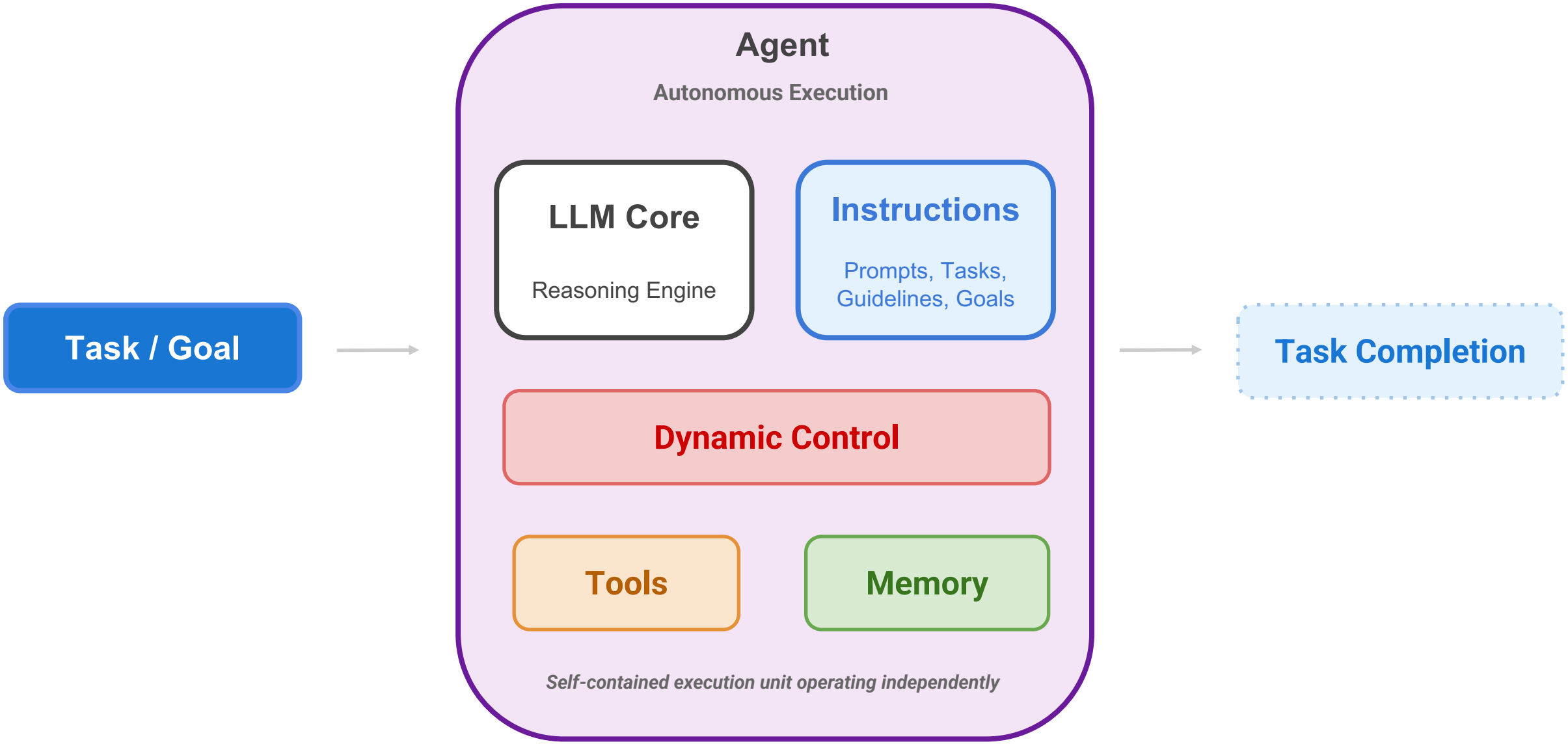
*"Agents are **systems** where LLMs dynamically direct their own processes and tool usage, keeping control over how a task gets done."* ~ Anthropic

*"Agent is a self-contained **execution unit** designed to act autonomously to achieve specific goals."* ~ Google

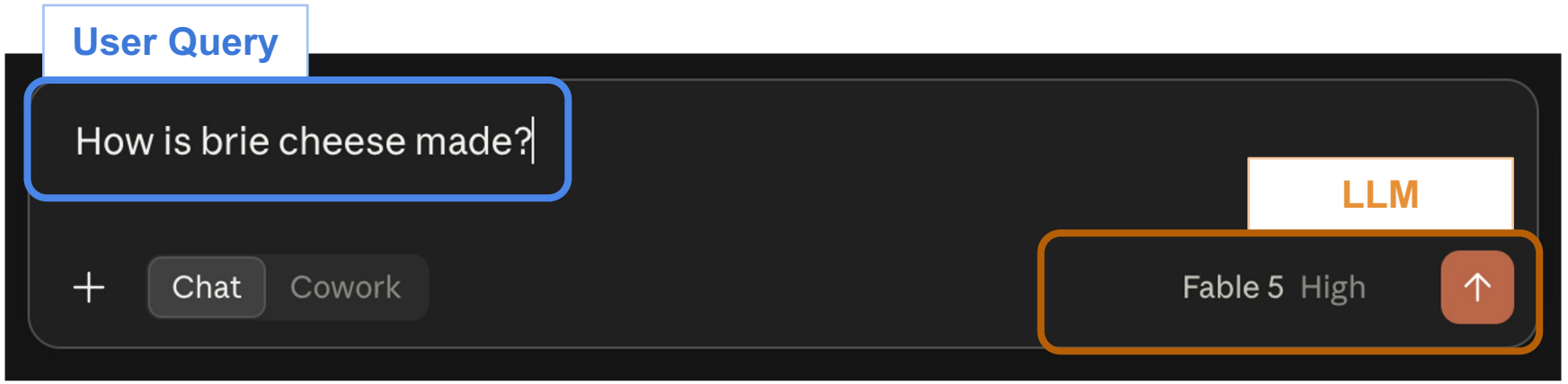
Agent Definition

Agent is a system where an LLM, given instructions and tools, runs autonomously until a task is done.

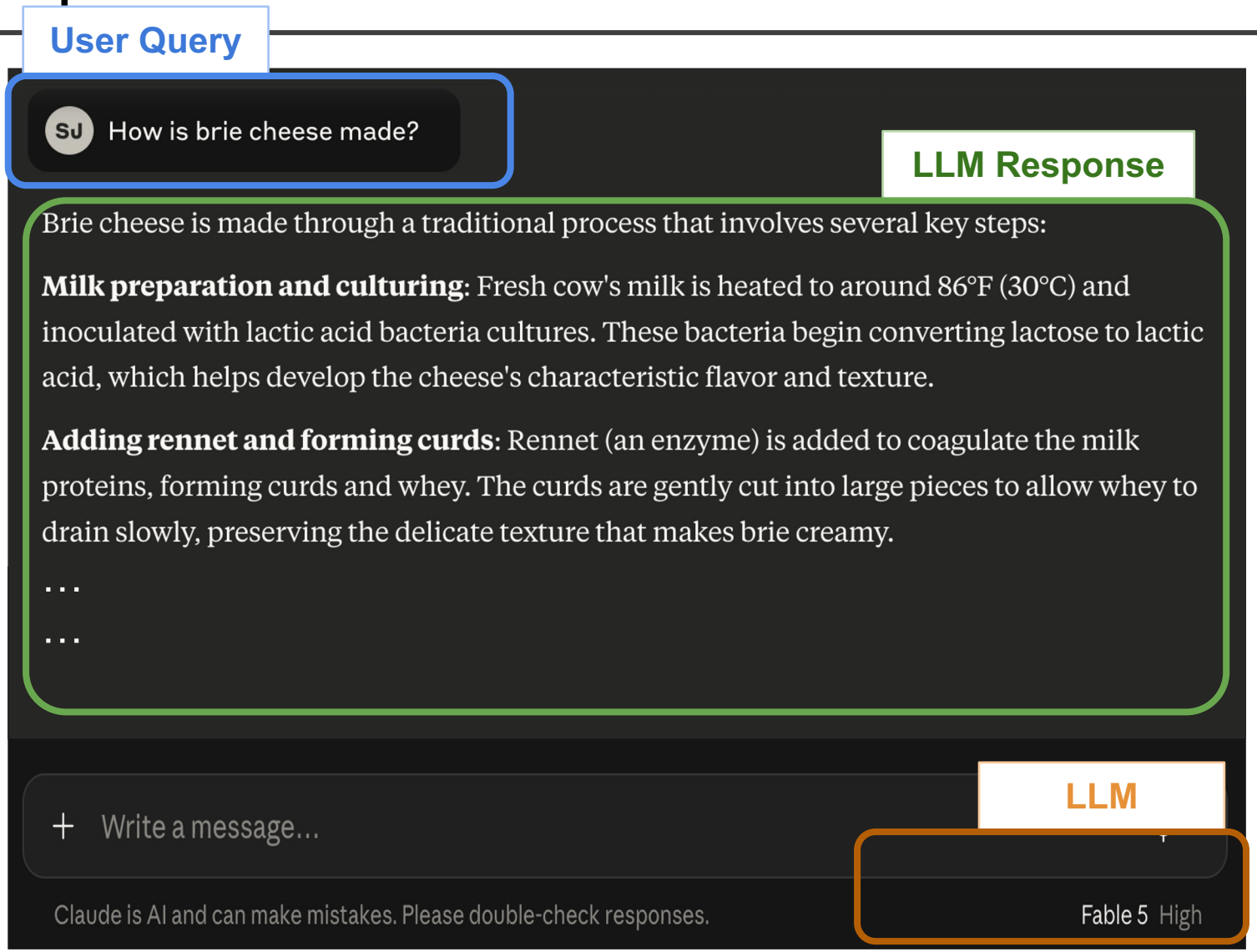
What is an Agent?



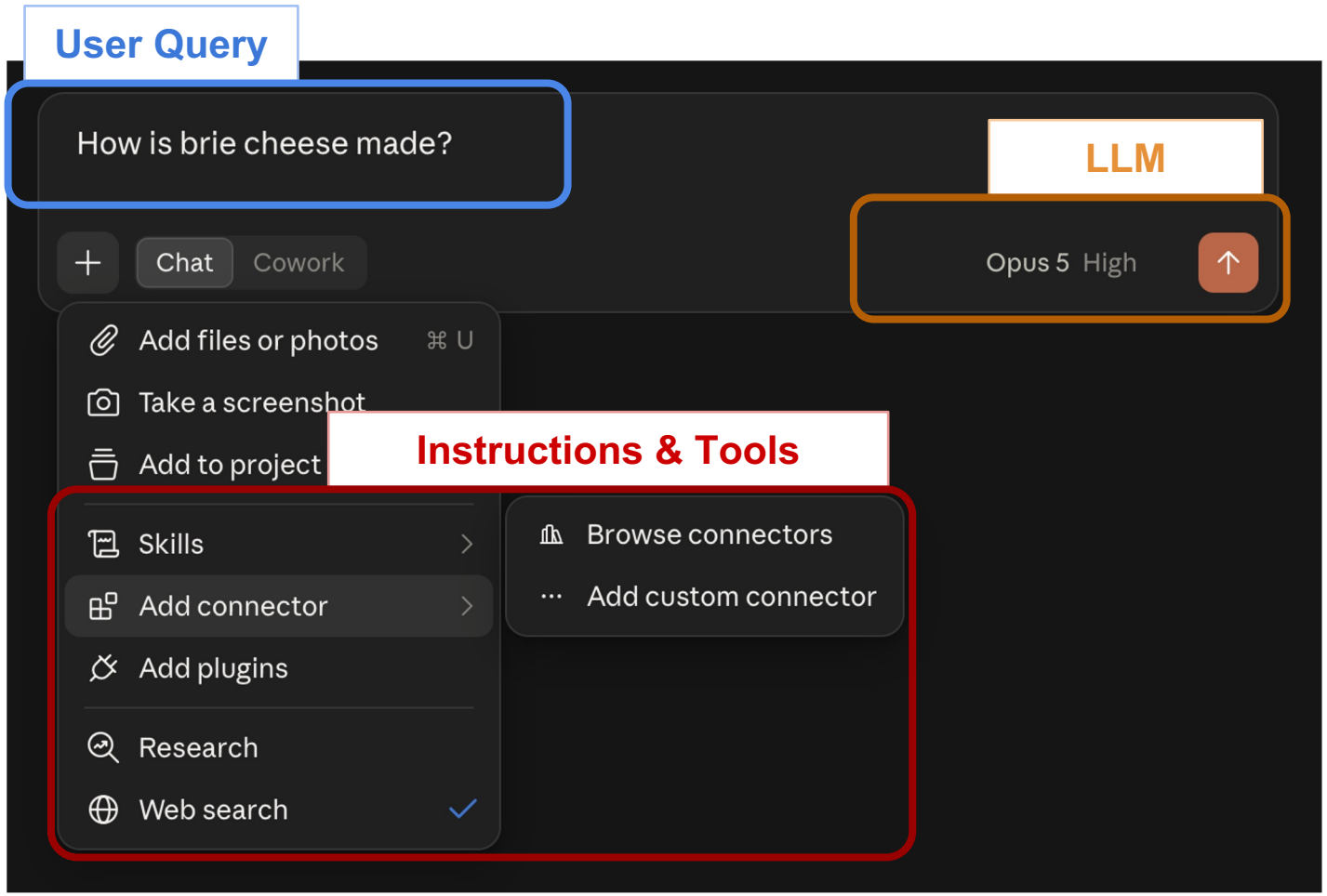
Agent Example: Claude



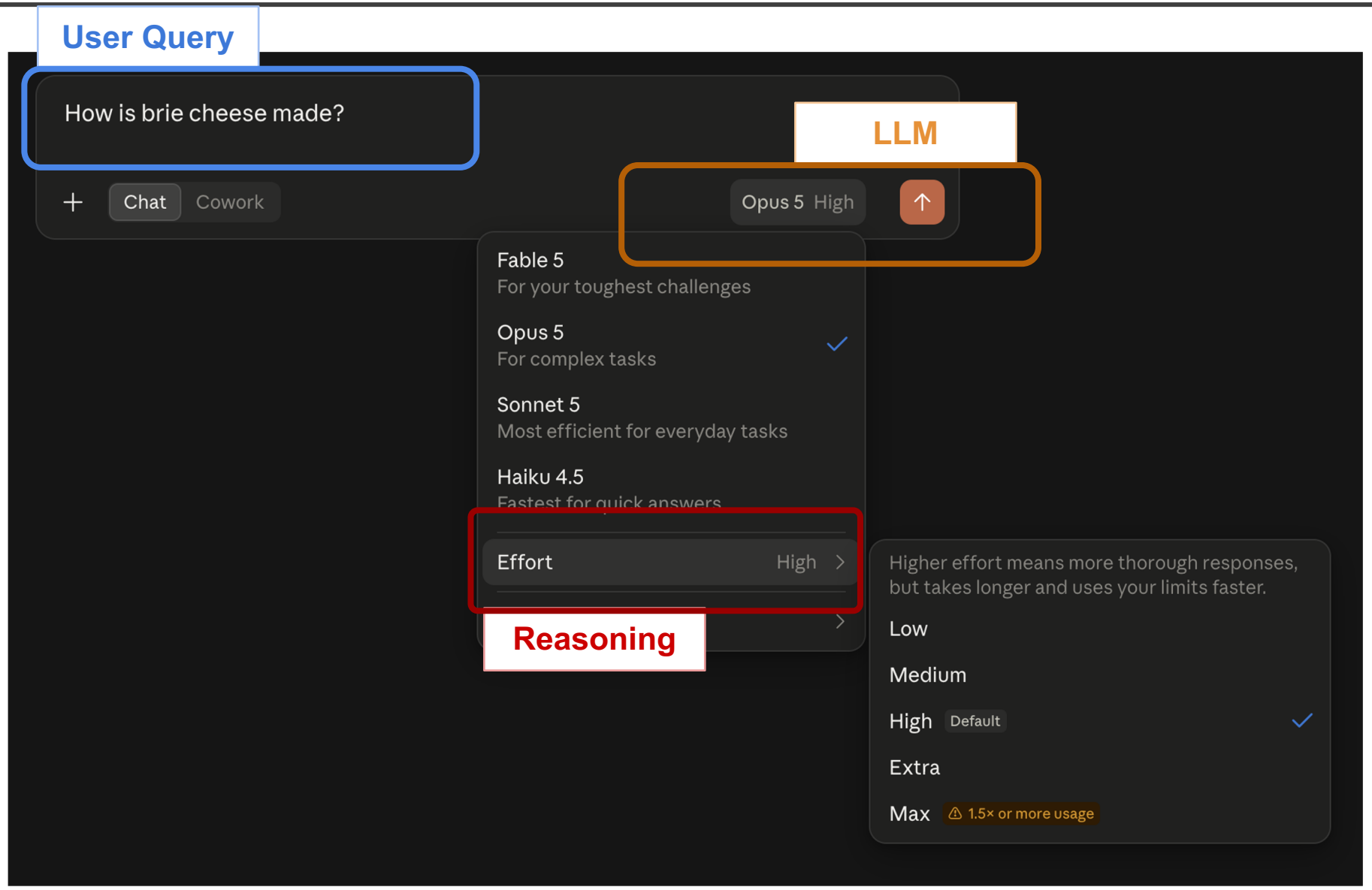
Agent Example: Claude



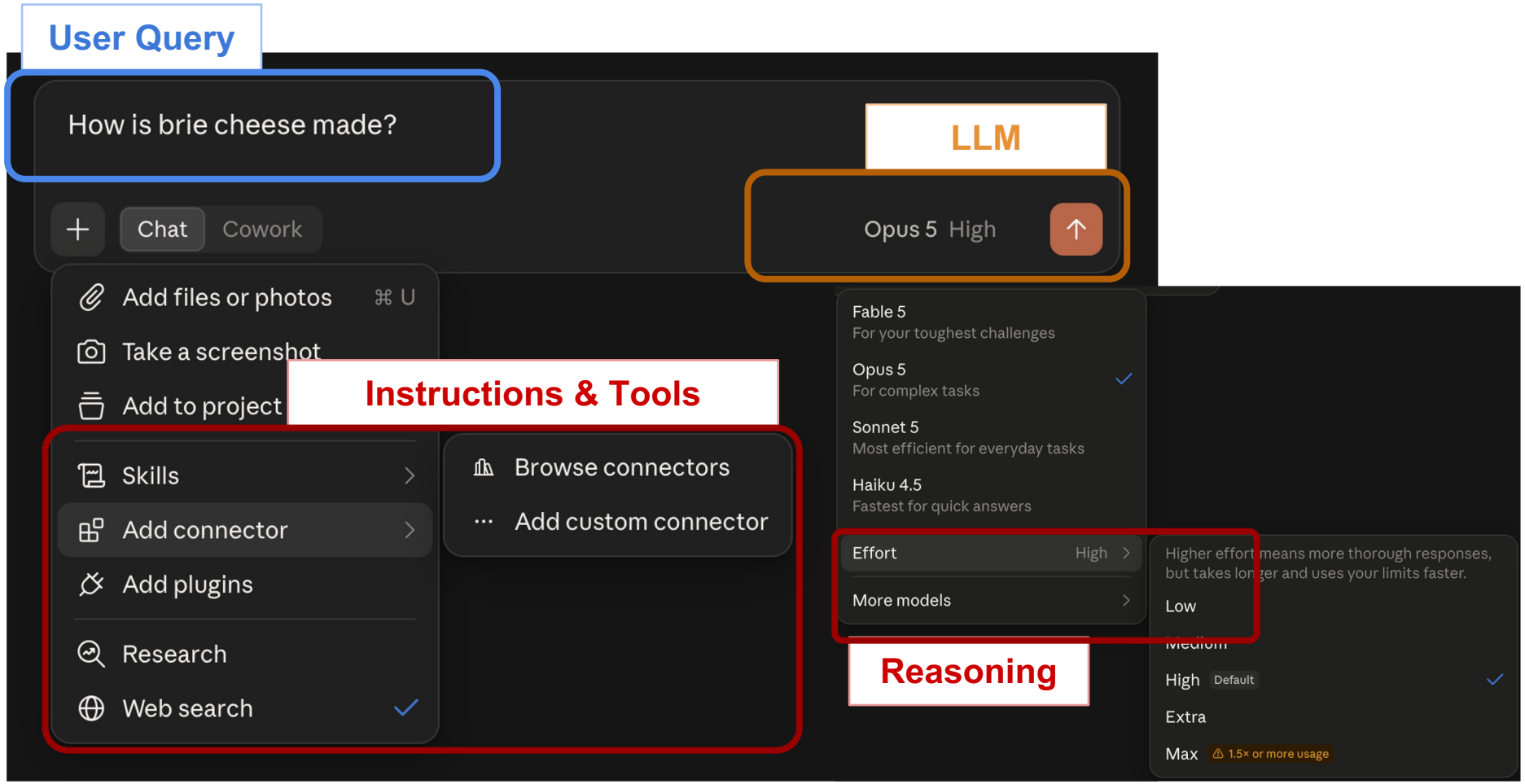
Agent Example: Claude



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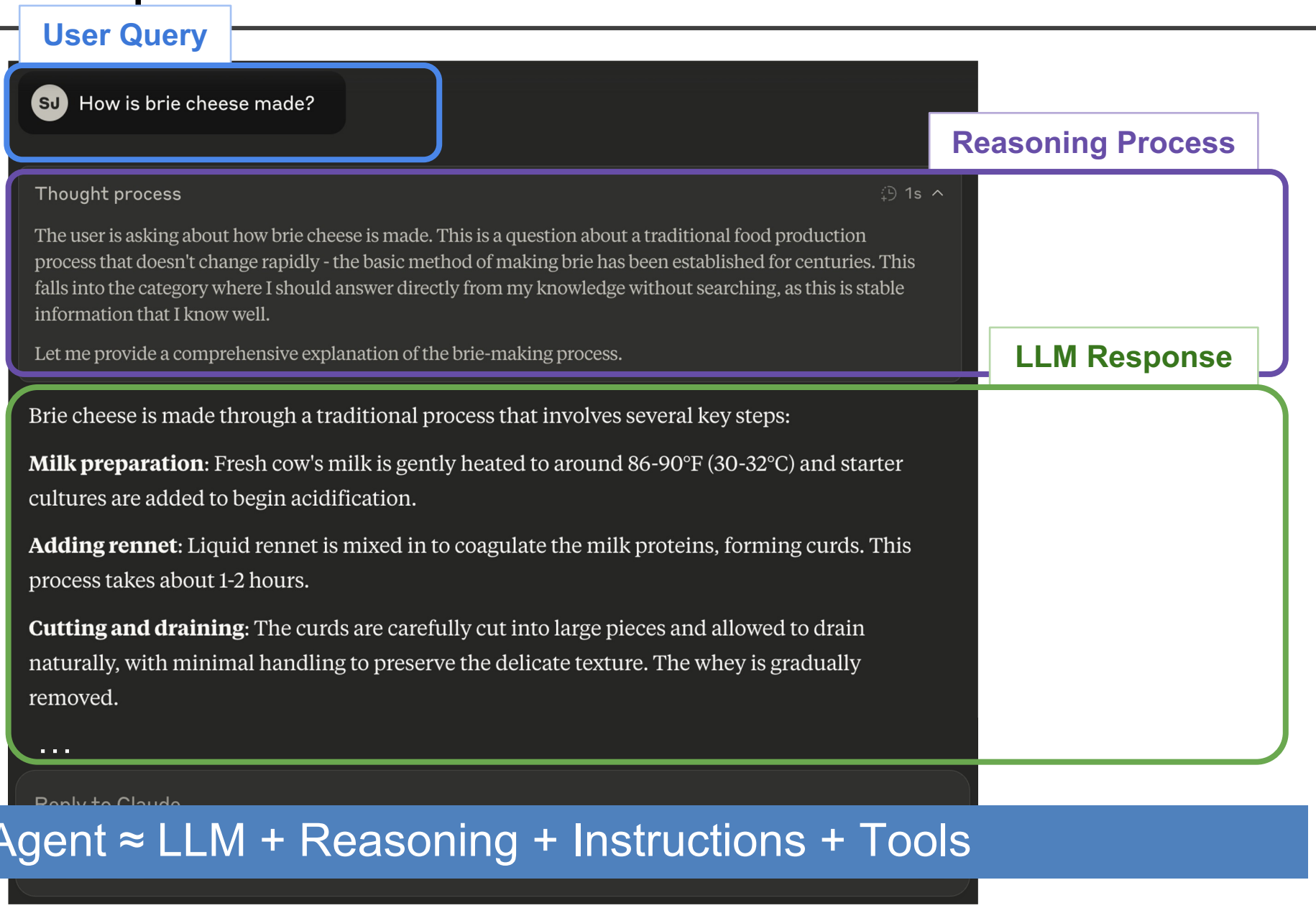


Agent Example: Claude



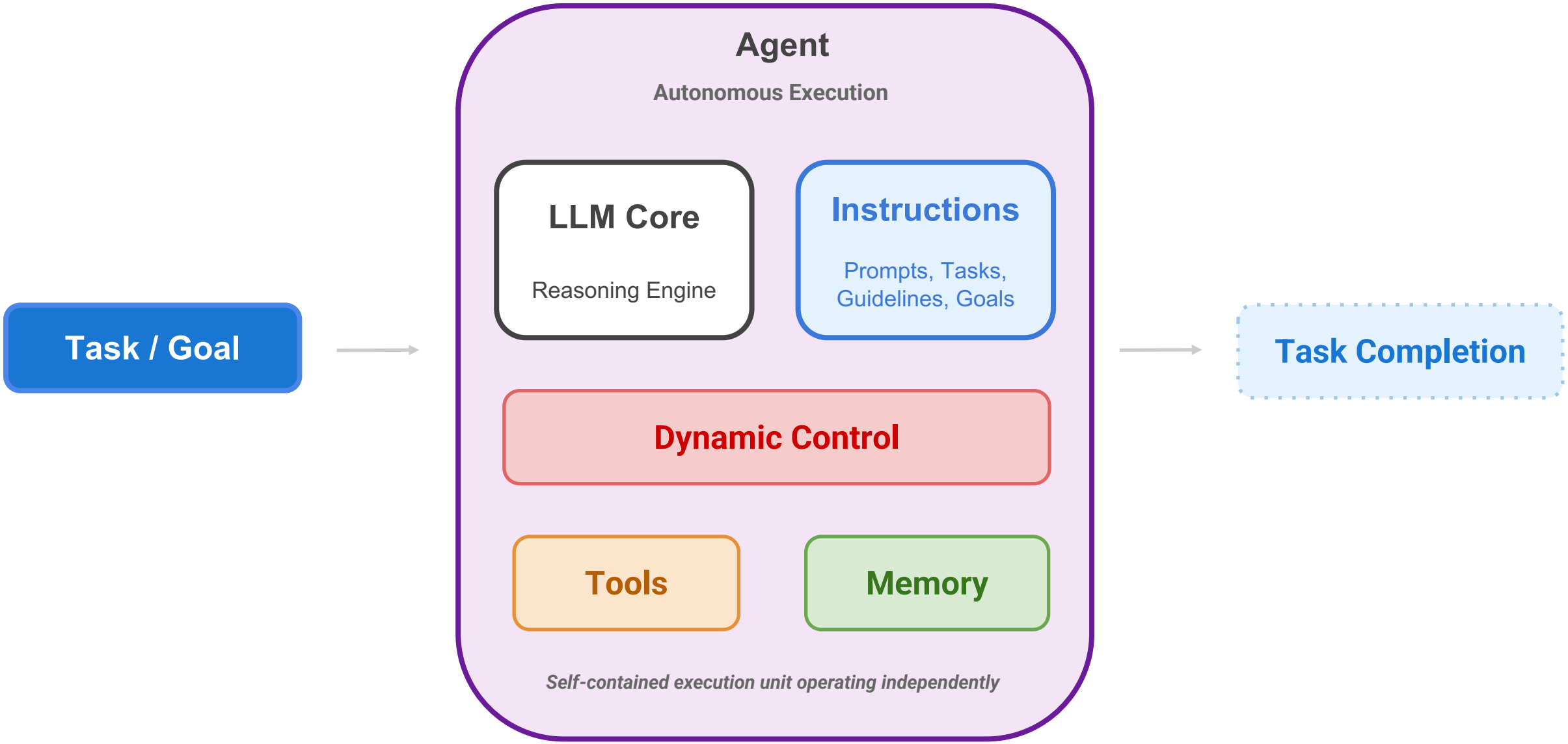
Agent ≈ LLM + Reasoning + Instructions + Tools

Agent Example: Claude



Agent ≈ LLM + Reasoning + Instructions + Tools

What is an Agent?



How an Agent works?

Step 1: LLM Core

- Serves as the reasoning engine
- Determines agents cognitive capabilities

- Key decisions:
  - Model selection (GPT, Claude, Gemini, etc)
  - Model parameters (temperature, top-p, max tokens)
  - Response format requirements

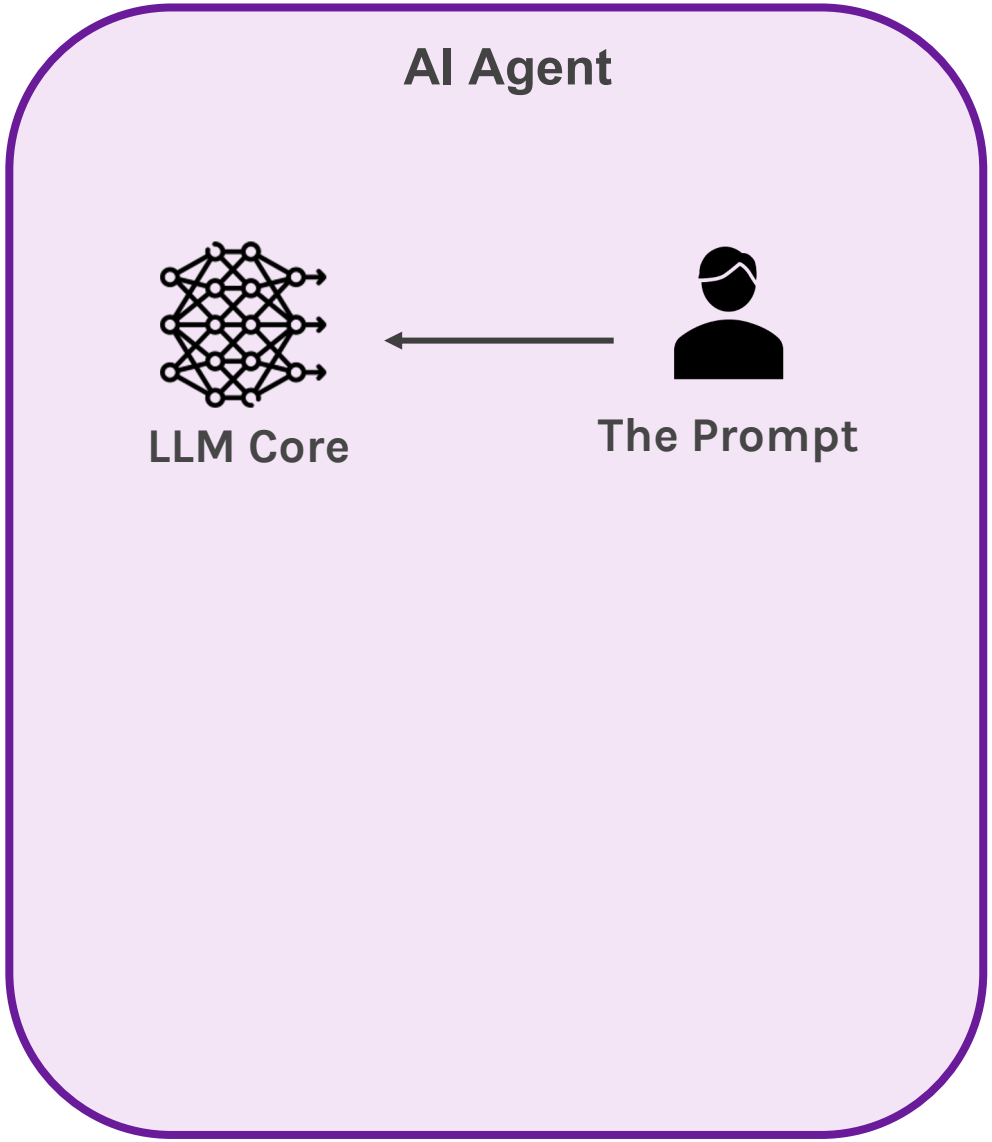


How an Agent works?

Step 2: Instructions

- Are prompts that instruct how the LLM behaves
- Configure the LLM core into a specific capabilities and focus.

- They define:
  - Identity & role
  - Guideline & Constraints
  - Goals & success criteria
  - Output formats

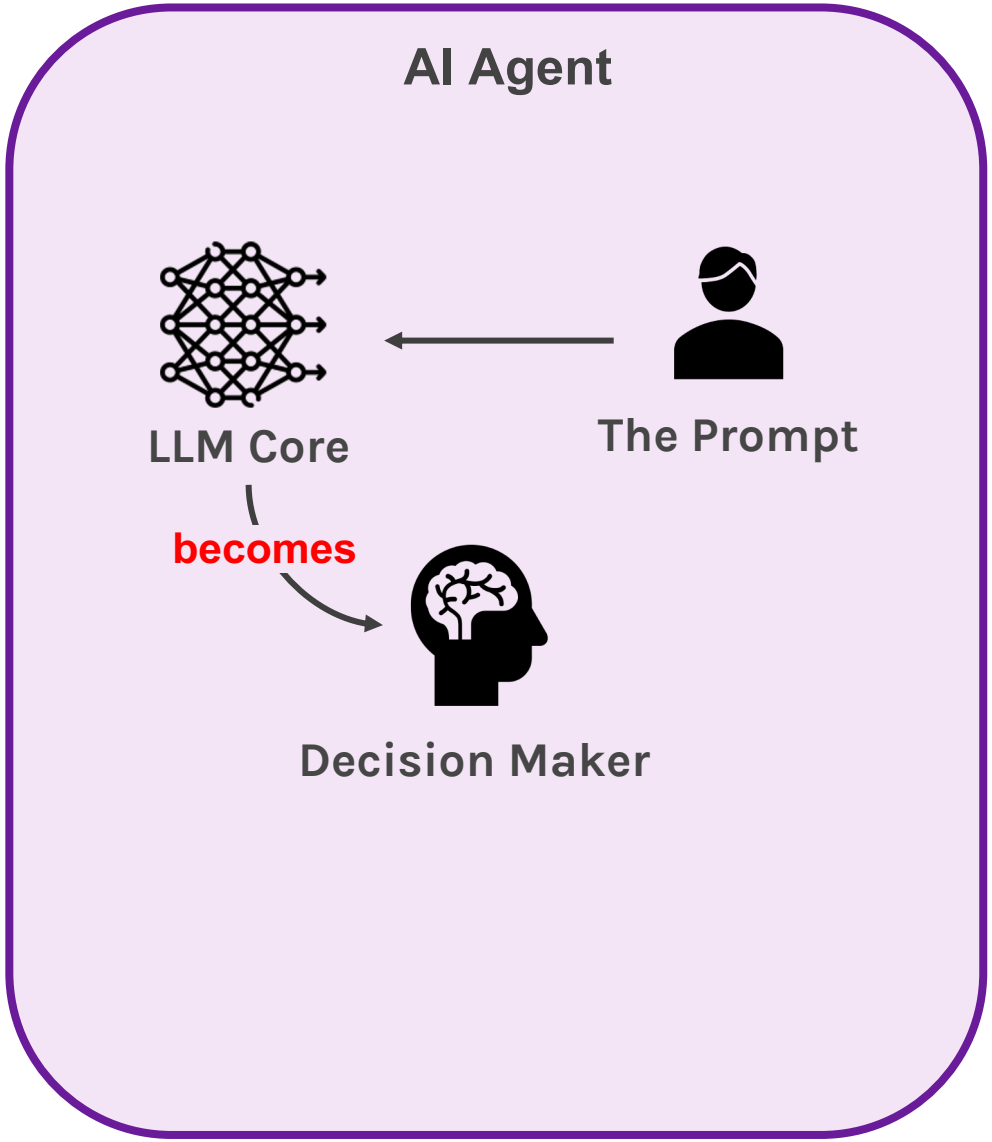


How an Agent works?

Step 3: Decision Maker / Dynamic Control

- Acts as the brain of the system
- Ensures all components are aligned and can process information to achieve goals

- Evaluates:
  - User intent & requirements
  - Available tools & resources
  - Constraints & guardrails
  - Context from memory

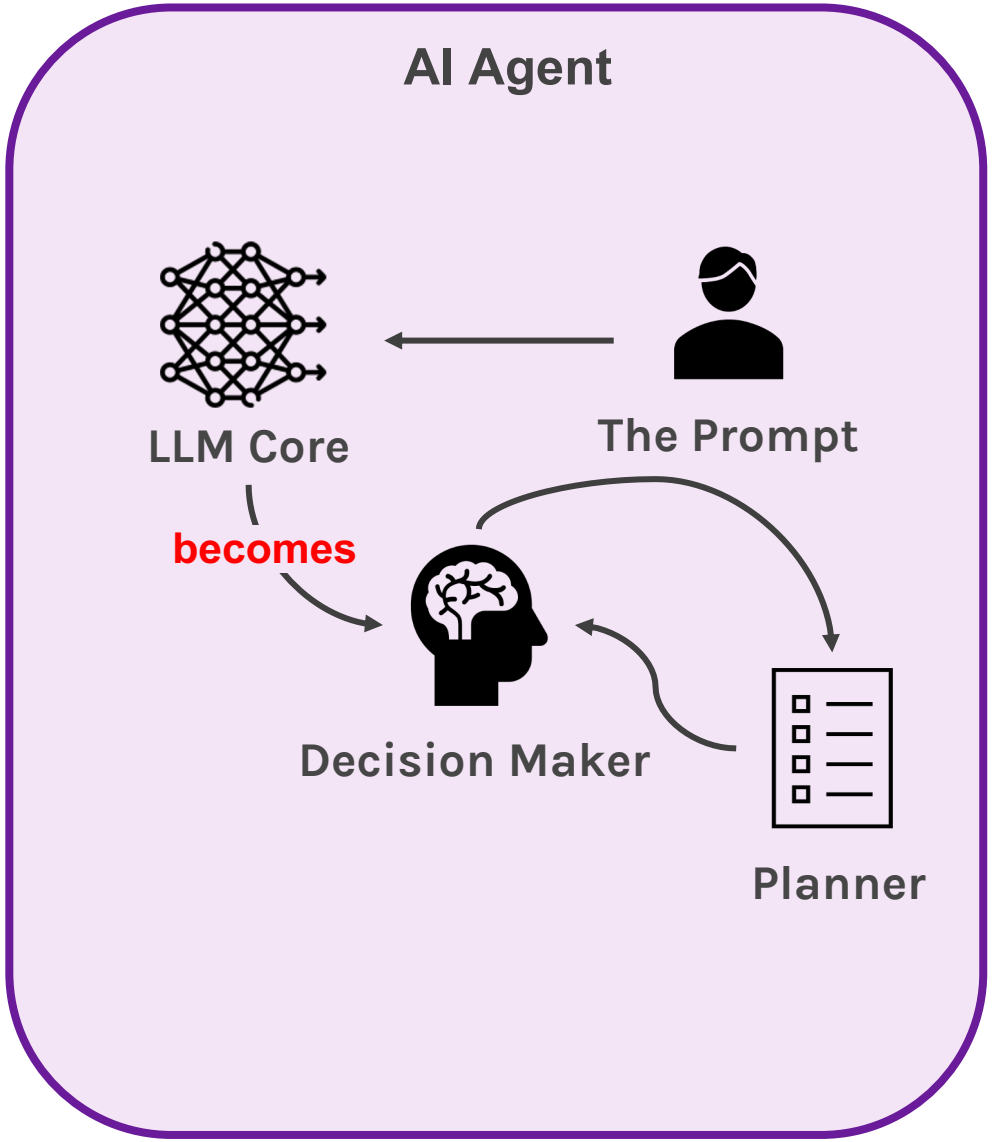


How an Agent works?

Step 4: Planner / Dynamic Control

- Determines HOW to accomplish the goal
- Enables emergent workflows to tackle novel problems by creating paths not explicitly covered in instructions

- Performs:
  - Breaks down task into executable steps
  - Sequences actions in logical order
  - Identify dependencies between tasks
  - Anticipate potential obstacles

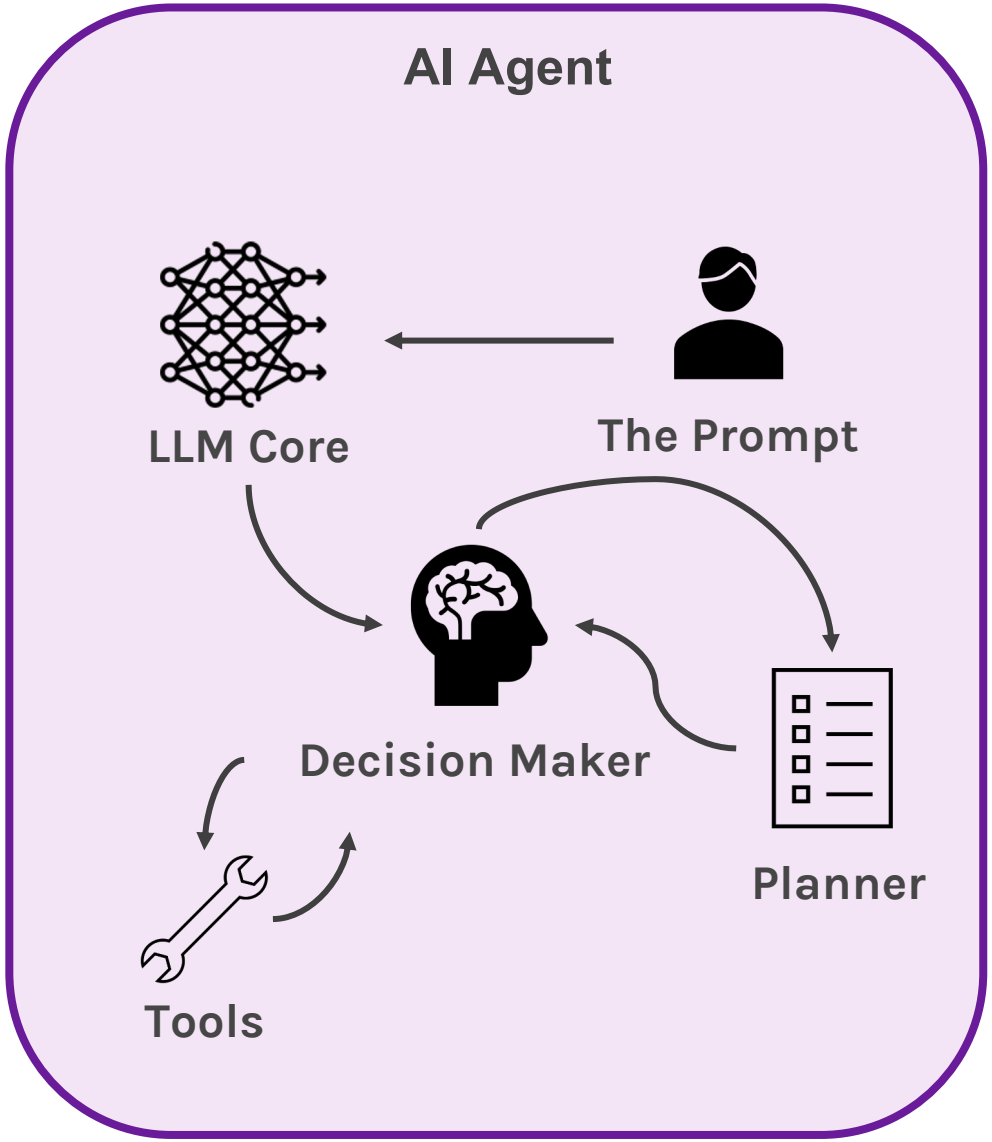


How an Agent works?

Step 5: Tools

- Are interfaces with the external world
- Transform agent from conversational to action-oriented system

- Must:
  - Include function calling (APIs, Databases, calculations, code execution, etc.)
  - Select appropriate tools for each task
  - Format and interest tool inputs and outputs

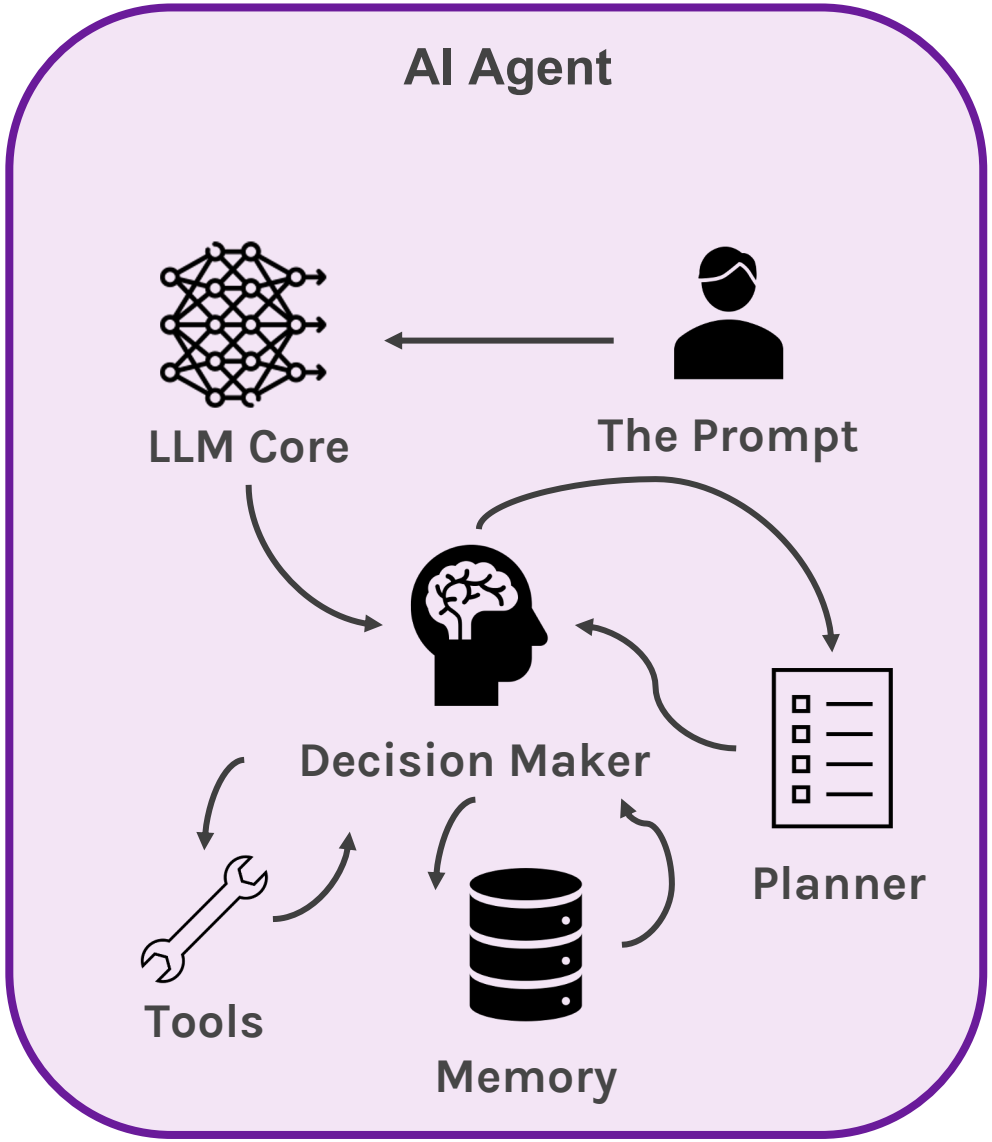


How an Agent works?

Step 6: Memory

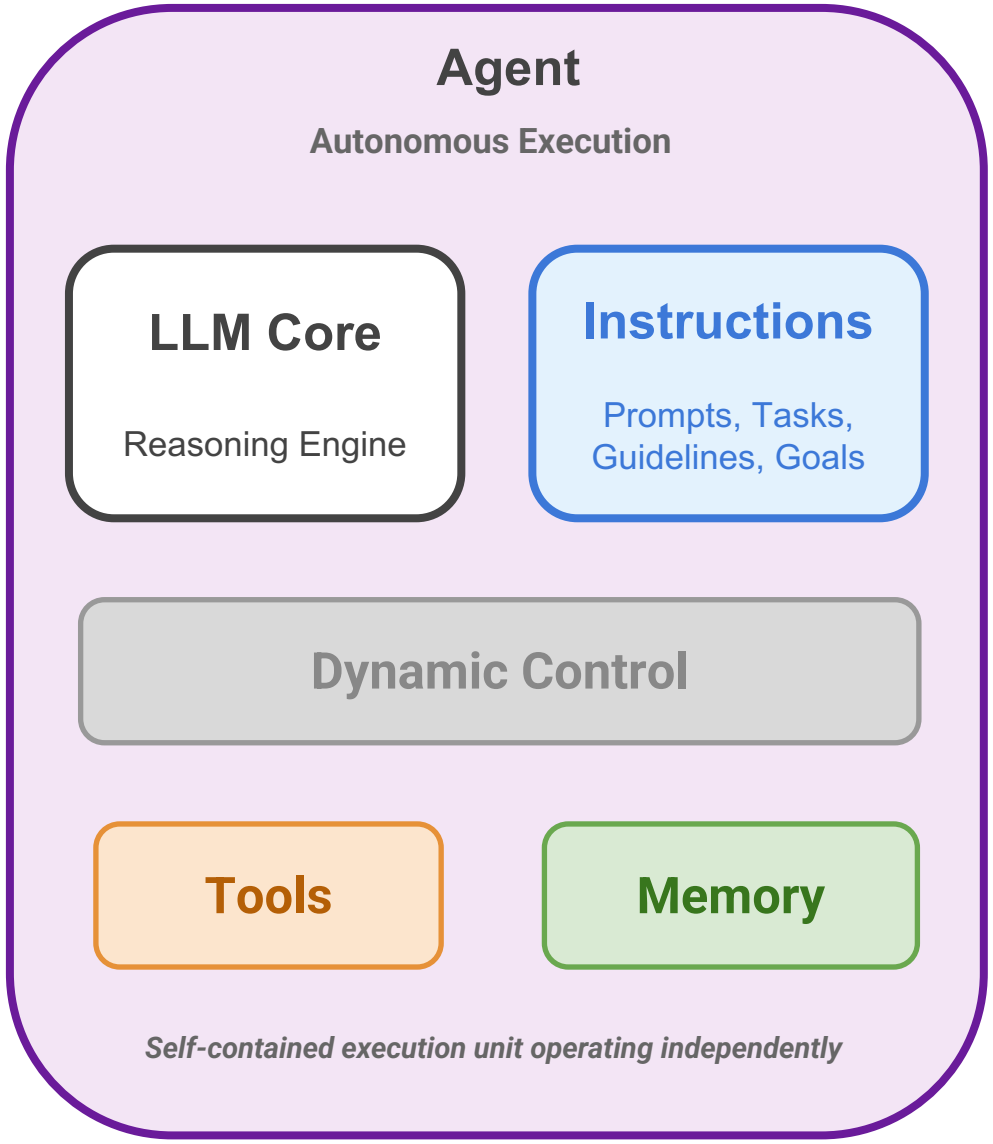
- Provides persistence and context over time
- Transforms isolated interactions into evolving relationships with user

- Used to:
  - Stores session history, user preferences, learned strategies
  - Enables continuity and personalization across multiple sessions
  - Enables self-monitoring over time



Agent: Building Blocks

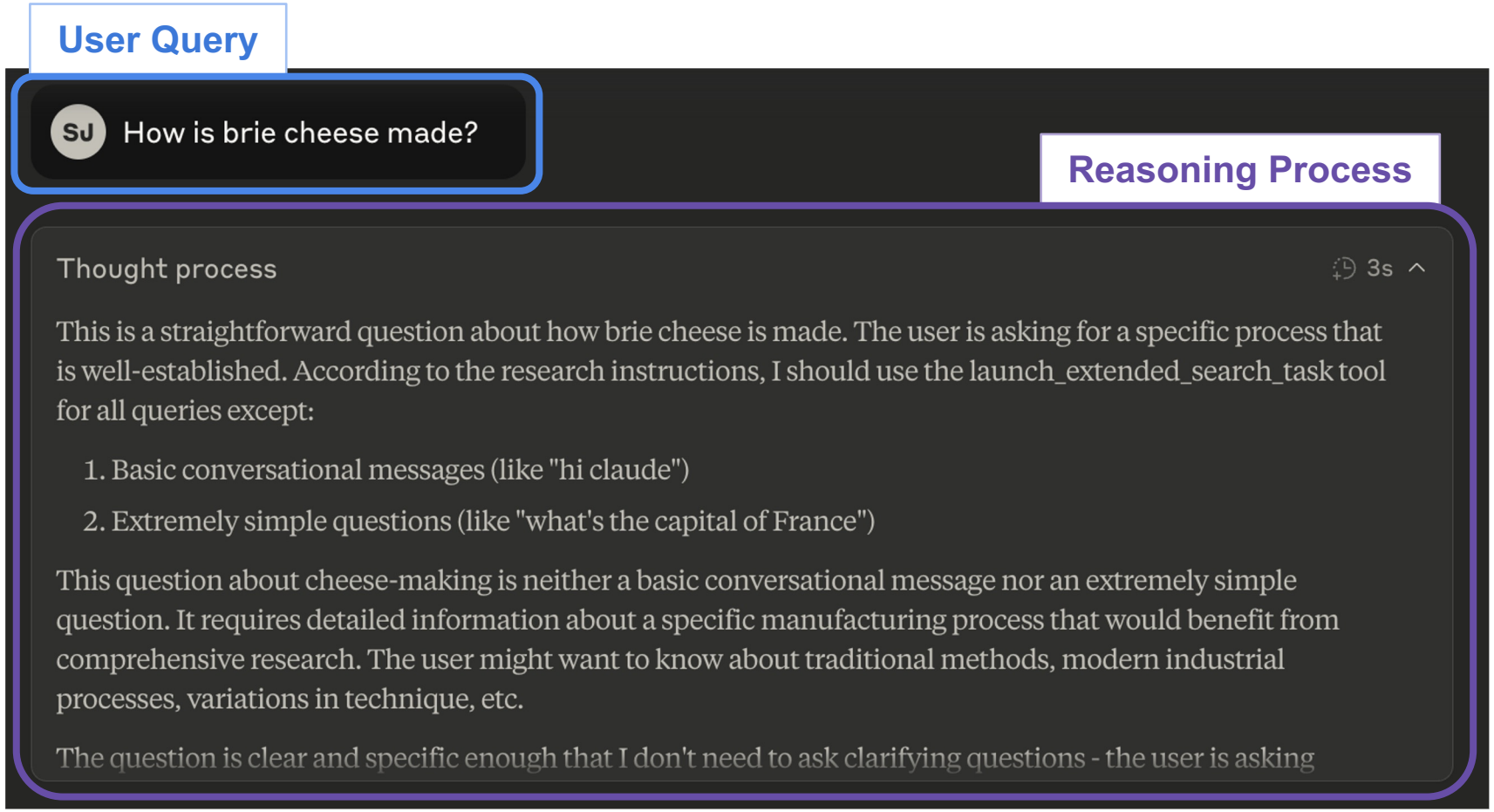
- **LLM Core**
- **Instructions**
  - System & User Prompts
  - Contexts
  - Output Schema
- **Tools**
  - Data
  - Actions
  - Orchestrations (flows)
- **Memory**
  - User sessions
  - User preferences



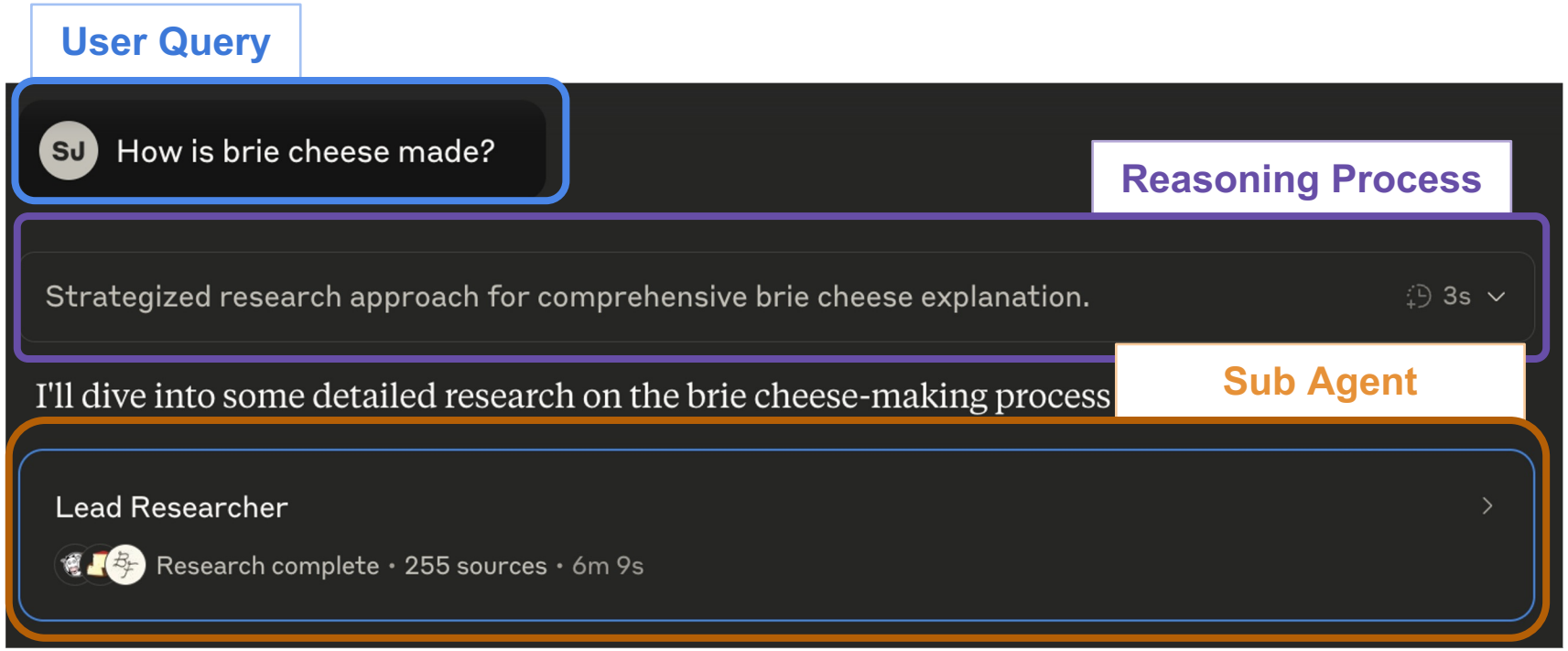
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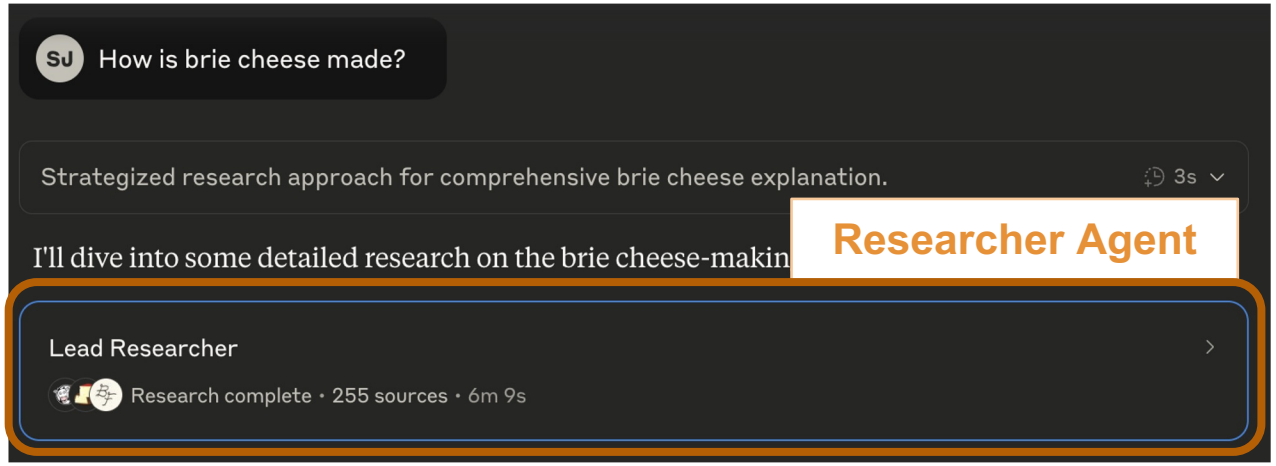
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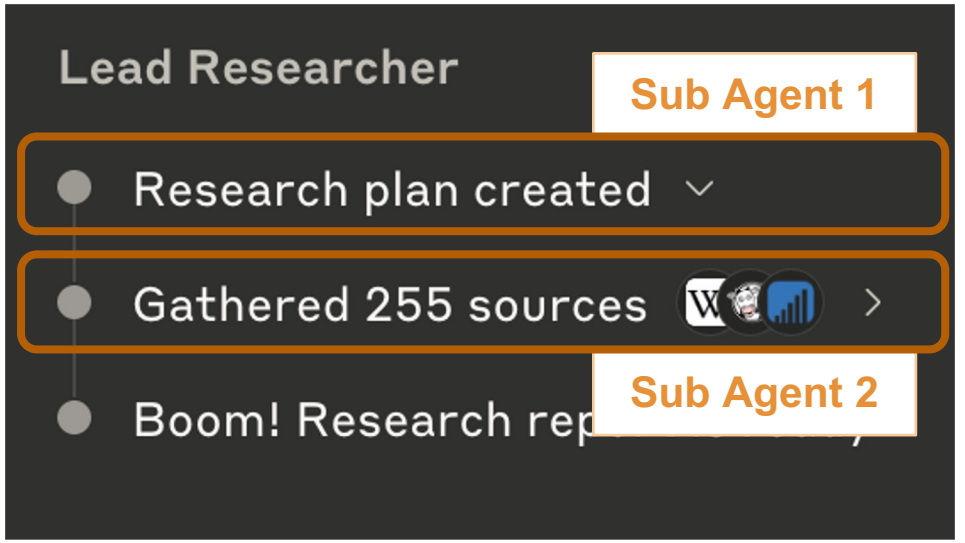


Agent Example: Claude



Agent Example: Claude





Agent Example: Claude

Research Planner Agent

Research plan created

The user is asking for a comprehensive explanation of how brie cheese is made. This is a detailed technical question about a specific food production process. Let me analyze this using the research process guidelines.

1. **Assessment and breakdown**:
2. **Query type determination**: This appears to be a **depth-first query** because:
3. **Detailed research plan development**: For this depth-first query, I should deploy multiple subagents to explore different aspects:

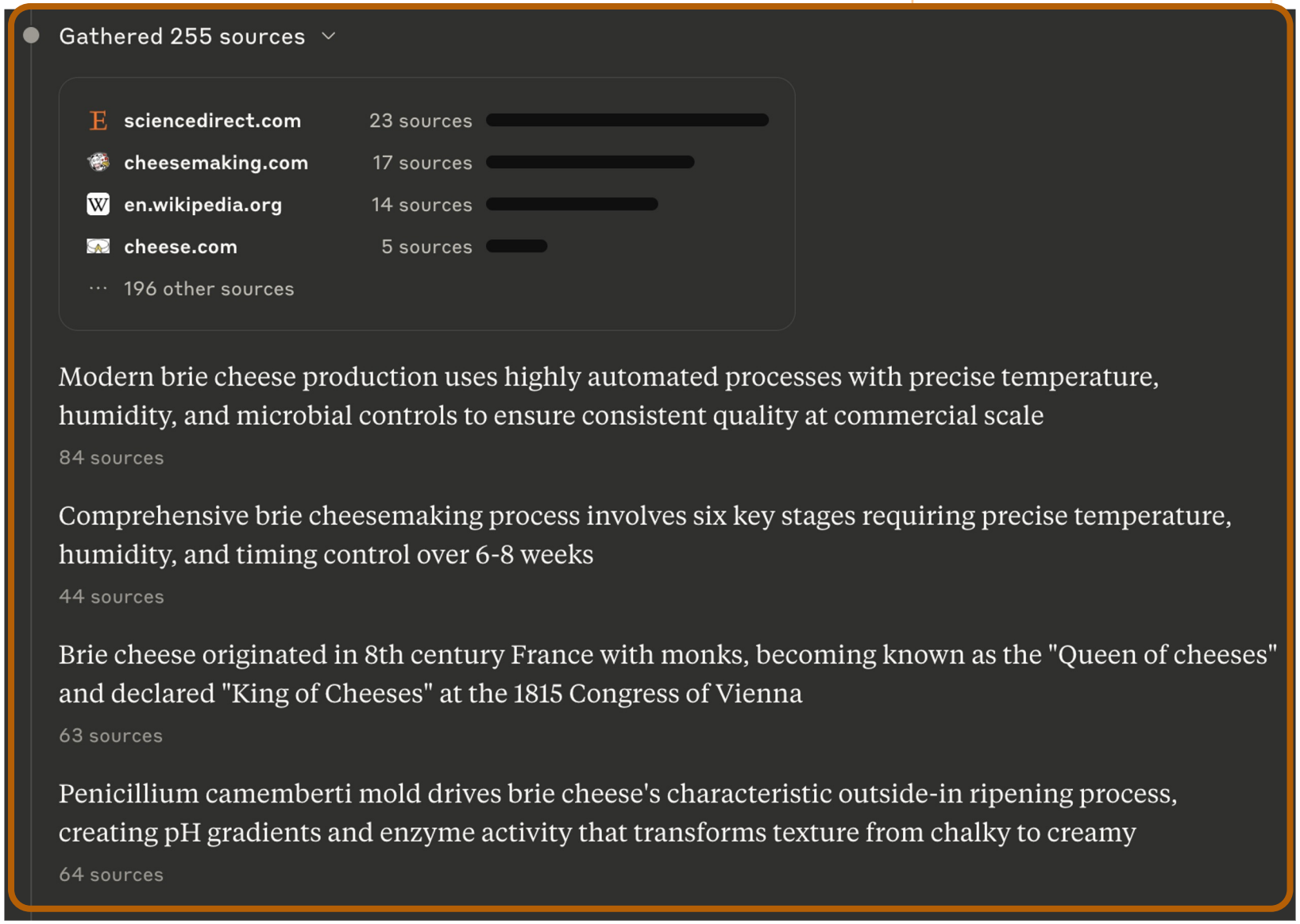
Instructions for more Sub Agents

I should create about 3-4 subagents to cover these different angles comprehensively.

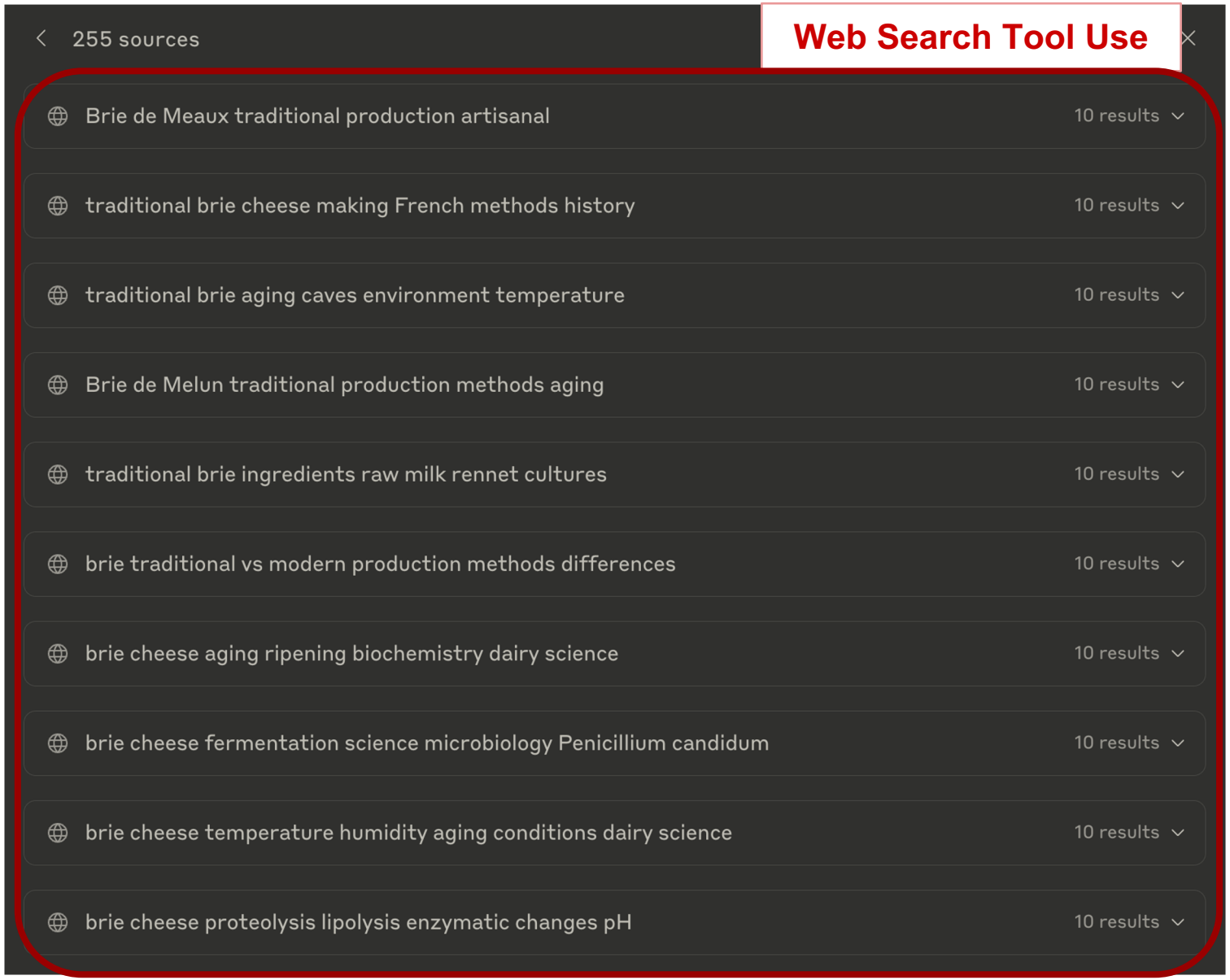
4. **Execution**: Deploy subagents in parallel to research different aspects, then synthesize into a comprehensive report.

Agent Example: Claude

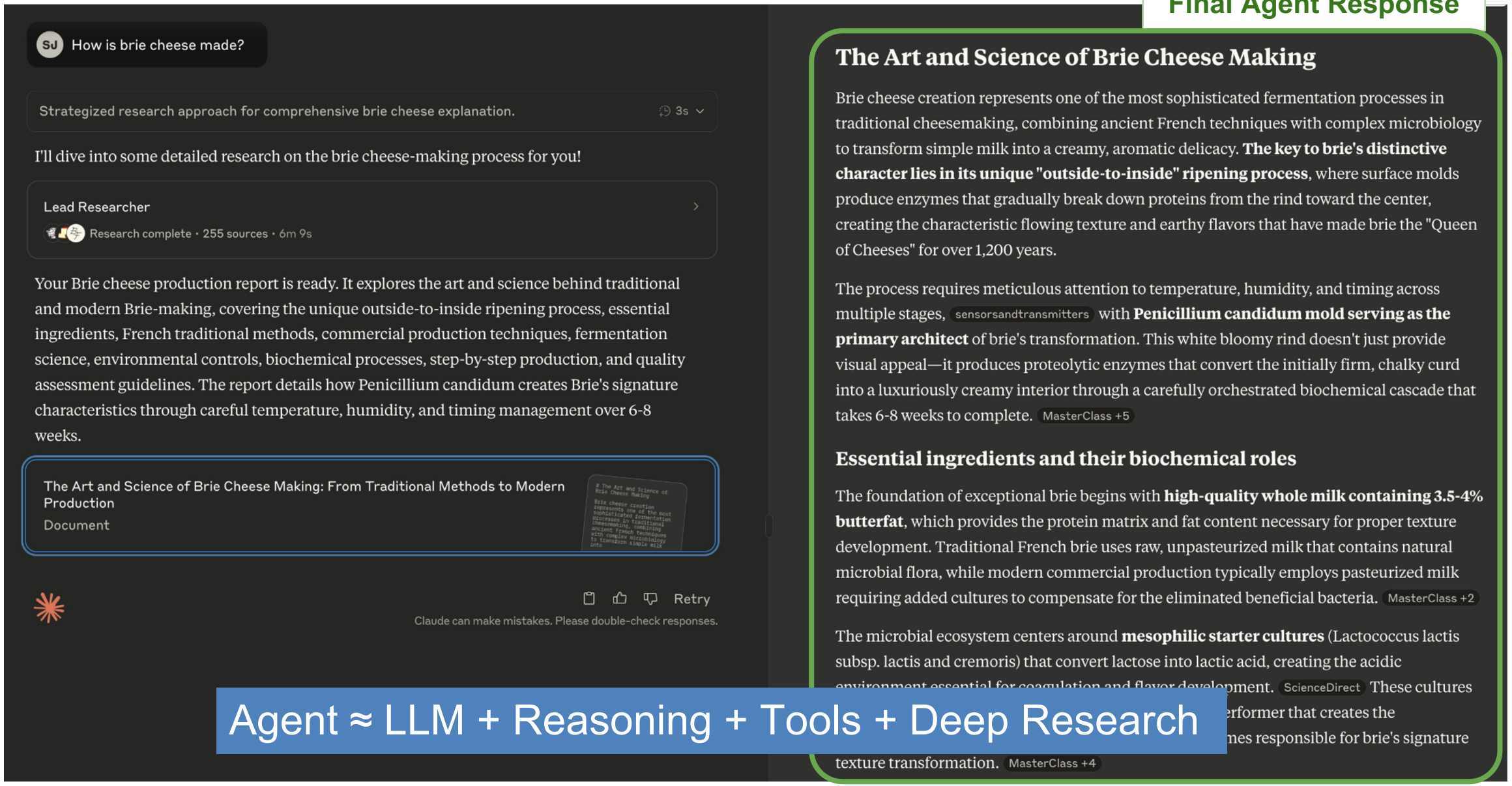
Web Search Agent



Agent Example: Claude



Agent Example: Claude



Final Agent Response

The Art and Science of Brie Cheese Making

Brie cheese creation represents one of the most sophisticated fermentation processes in traditional cheesemaking, combining ancient French techniques with complex microbiology to transform simple milk into a creamy, aromatic delicacy. **The key to brie's distinctive character lies in its unique "outside-to-inside" ripening process**, where surface molds produce enzymes that gradually break down proteins from the rind toward the center, creating the characteristic flowing texture and earthy flavors that have made brie the "Queen of Cheeses" for over 1,200 years.

The process requires meticulous attention to temperature, humidity, and timing across multiple stages, sensorsandtransmitters with **Penicillium candidum mold serving as the primary architect** of brie's transformation. This white bloomy rind doesn't just provide visual appeal—it produces proteolytic enzymes that convert the initially firm, chalky curd into a luxuriously creamy interior through a carefully orchestrated biochemical cascade that takes 6-8 weeks to complete. MasterClass +5

Essential ingredients and their biochemical roles

The foundation of exceptional brie begins with **high-quality whole milk containing 3.5-4% butterfat**, which provides the protein matrix and fat content necessary for proper texture development. Traditional French brie uses raw, unpasteurized milk that contains natural microbial flora, while modern commercial production typically employs pasteurized milk requiring added cultures to compensate for the eliminated beneficial bacteria. MasterClass +2

The microbial ecosystem centers around **mesophilic starter cultures** (Lactococcus lactis subsp. lactis and cremoris) that convert lactose into lactic acid, creating the acidic environment essential for coagulation and flavor development. ScienceDirect These cultures erformer that creates the mes responsible for brie's signature texture transformation. MasterClass +4

Agent ≈ LLM + Reasoning + Tools + Deep Research

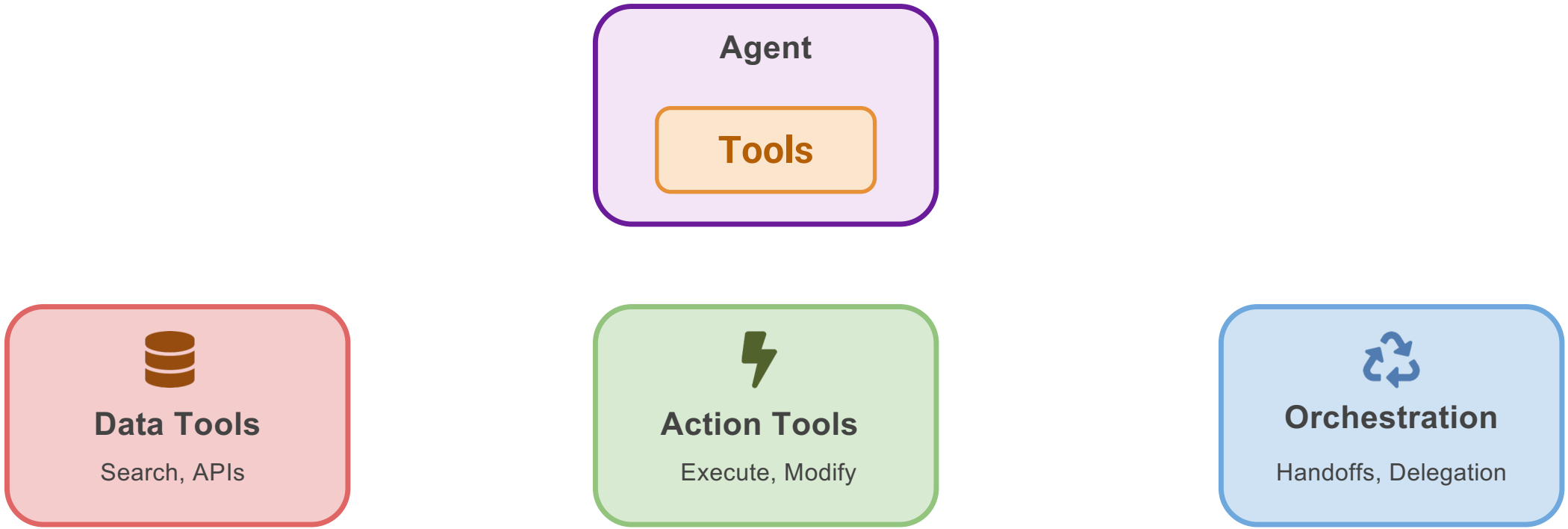
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6. Tutorials

Tools

Tools:

- The Agent's interface to the external world.
- They transform agents from conversational to action-oriented systems
- Are functions with defined schemas that agents can invoked autonomously.



Tools

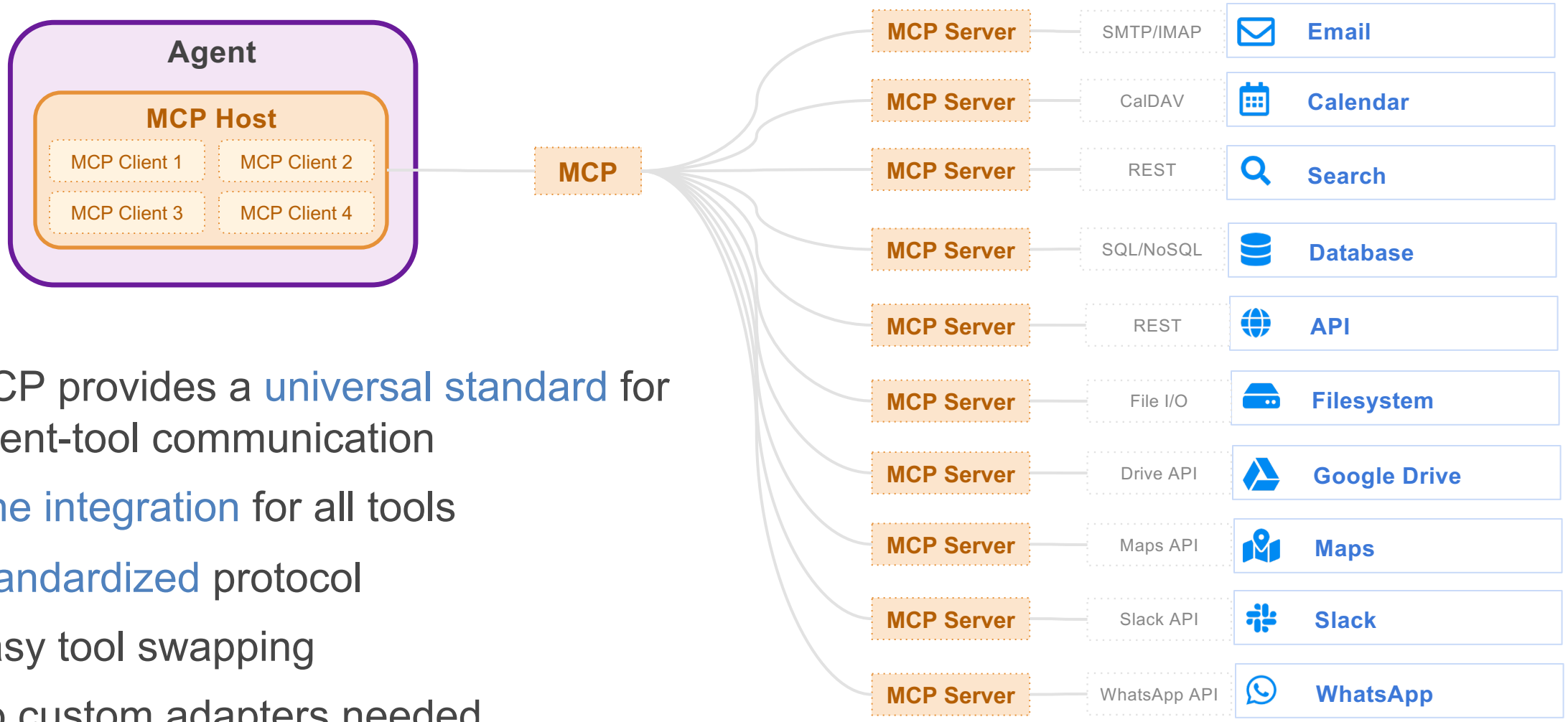


Agent tools can connect to any external system.

Do you see a problem?

- Every tool has different integrations / protocols
- Swapping tools require a lot of work

Model Context Protocol (MCP)



- MCP provides a universal standard for agent-tool communication
- One integration for all tools
- Standardized protocol
- Easy tool swapping
- No custom adapters needed

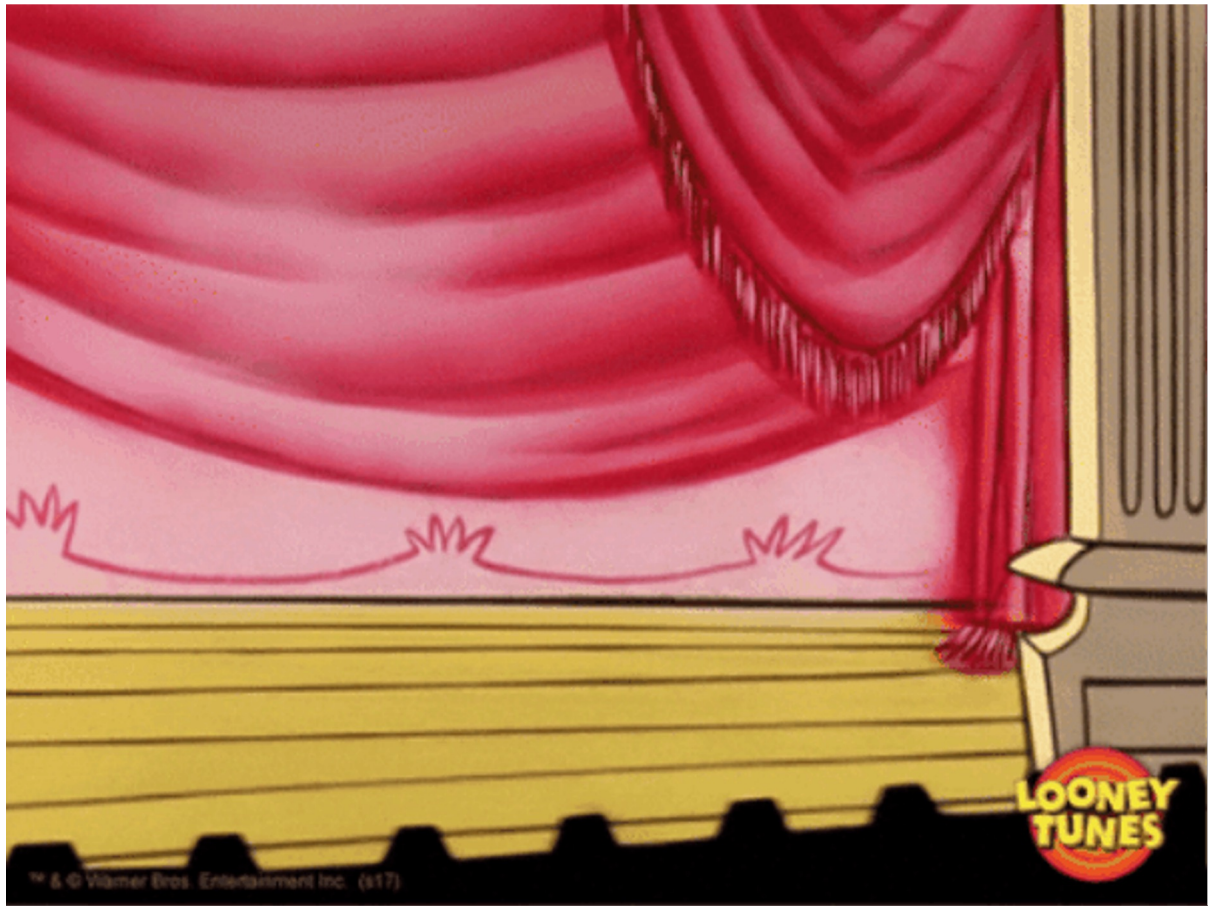
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Tutorial: Agents

Tutorials

- Cheese Quiz (Notebook): https://colab.research.google.com/drive/14UKW-Z7kFPuCwVqyUFcLxeA1KF70fwfm?usp=sharing
- lm-rag with agents (Github): https://github.com/dlops-io/llm-rag?tab=readme-ov-file#agents
- Travel Planner (Notebook): https://colab.research.google.com/drive/1I_jEHJsVkGxNoSC1c4wEoVF2QL6vCVY?usp=sharing
- Cheese Newsletter (Notebook): https://colab.research.google.com/drive/1nnHrJ4Si3tpgUQl6HqlcwxKymV1-ecPv?usp=sharing



THANK YOU